

Quantization-Aware and Tensor-Compressed Training of Transformers for Natural Language Understanding

Zi Yang¹, Samridhi Choudhary², Siegfried Kunzmann², Zheng Zhang¹

¹ Department of Electrical & Computer Engineering, University of California, Santa Barbara, CA

² Amazon Alexa AI

ziy@ucsb.edu, samridhc@amazon.com, kunzman@amazon.com, zhengzhang@ece.ucsb.edu

Abstract

Fine-tuned transformer models have shown superior performances in many natural language tasks. However, the large model size prohibits deploying high-performance transformer models on resource-constrained devices. This paper proposes a quantization-aware tensor-compressed training approach to reduce the model size, arithmetic operations, and ultimately runtime latency of transformer-based models. We compress the embedding and linear layers of transformers into small low-rank tensor cores, which significantly reduces model parameters. A quantization-aware training with learnable scale factors is used to further obtain low-precision representations of the tensor-compressed models. The developed approach can be used for both end-to-end training and distillation-based training. To improve the convergence, a layer-by-layer distillation is applied to distill a quantized and tensor-compressed student model from a pre-trained transformer. The performance is demonstrated in two natural language understanding tasks, showing up to $63\times$ compression ratio, little accuracy loss and remarkable inference and training speedup.

Index Terms: model compression, tensor decomposition, quantization, natural language understanding

1. Introduction

Transformer models [1] have been widely used for natural language understanding (NLU) [2–4] and automatic speech recognition (ASR) [5–7]. Typically, larger pre-trained transformer models perform better on downstream tasks [2, 8–10]. However, these large-size models cannot be deployed directly on edge devices due to the limited computing, memory, and energy resources as well as low latency requirement. As a result, model compression has become an indispensable step to enable efficient deployment of large NLU and ASR models on resource-constrained hardware platforms [11–13]. Existing works have studied NLU and ASR model compression via knowledge distillation [14–17], quantization [18–20] and low-rank matrix factorization [21, 22]. Among these approaches, low-rank matrix compression normally achieves much higher compression ratios.

Meanwhile, studies in the applied math community have shown that tensor decomposition [23] often achieves a much higher compression ratio than matrix compression approaches. As a high-dimensional generalization of matrix decompositions, low-rank tensor decomposition has achieved state-of-the-art results in neural network compression [24–27], including both post-training compression and end-to-end compressed training. Recently, tensor decomposition has also been employed to compress transformer models used in natural language modeling [28]. Since many edge devices (e.g., embed-

ded CPU, embedded GPU and FPGA) support low-precision computation, it is natural to ask if low-precision tensor compression can be used to achieve further cost reduction on edge devices. A previous study [29] investigated low-precision training of tensor-compressed models, but it shows that directly applying existing low-precision training in the tensor-compressed setting can cause a remarkable accuracy drop even on a simple two-layer perceptron network.

In this work, we present a quantization-aware and tensor-compressed training approach for transformers. We first use low-rank tensor train (TT) and tensor-train matrix (TTM) formats to represent the embedding tables and linear layers respectively, which achieve significant parameter reduction. To further reduce memory and computing costs, we apply quantization-aware training with learnable scale factors, which enforces the low-rank tensor factors of transformer models into low precision. Our work uses 2-, 4-, or 8-bit fixed-point uniform quantization. The proposed quantization-aware and tensor-compressed training can be used for both end-to-end training and post-training compression. In order to leverage the information of pre-trained transformer models to save the training cost, we further employ layer-by-layer distillation [15] to match the internal outputs and attention probabilities of the original model and our low-precision tensor-compressed model to maintain the generalization capability. This layer-by-layer distillation can avoid the divergence issue of distilling all layers in a tensor-compressed format. We demonstrate the quantization-aware and tensor-compressed training approach on NLU tasks, ATIS dataset [30] and GLUE benchmark [31]. We perform end-to-end training on the ATIS dataset and compress the BERT-base via layer-by-layer distillation on the GLUE benchmark. In both tasks, our approach reaches ultra-low model size with little performance degradation.

2. Methodology

2.1. Tensor-Compressed Transformer Training

A typical transformer model [1] consists of an embedding table and a set of encoder blocks, where each encoder has one self-attention layer and one feed-forward layer. All self-attention and feed-forward layers are composed of linear layers. The embedding table can be regarded as a special type of linear layer. Tensor-compressed transformer compresses the weight matrices of the linear layers into small tensor cores. We directly train the small tensor cores rather than larger weight matrices.

Consider the linear layer $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$, where $\mathbf{x} \in \mathbb{R}^N$ is the input, $\mathbf{W} \in \mathbb{R}^{M \times N}$ is the weight matrix, and $\mathbf{b} \in \mathbb{R}^M$ is the bias vector. The weight $\mathbf{W}$ is reshaped into a tensor $\mathcal{W} \in \mathbb{R}^{m_1 \times \dots \times m_d \times n_1 \times \dots \times n_d}$, where $\prod_{i=1}^d m_i = M$ and $\prod_{i=1}^d n_i = N$. Then we employ either the tensor-train (TT) for-